TIMPANOGOS CAVE

Design Document

2022-2023

NATIONAL PARK SERVICE

Design Specifications

1

1.0 Strategy

Our strategy based on our client Cami and our target audience is to design trail interactive experiences that engage hikers and have them leave Timpanogos Cave feeling a greater understanding about the cave and its history, an appreciation for nature, or a needed connection with oneself.

Characteristics and Goals of target audience

- Broad demographic
- Allows us to know even more that we need good onboarding
- Frequent Timpanogos Cave hikers as well as tourists and first time vistors
- Physical impairments that may affect users experience
- Leave hikers feeling something good afterwards

Although we have a broad audience, our primary persona is going to be hikers that are new to Timpanogos Cave or do not hike the trail frequently (years in between hiking it).

This is our primary persona because frequent hikers are there likely for the exercise rather than learning more about the cave and will not likely use the QR codes. New hikers, or hikers that are not frequent to Timpanogos Cave will be more likely to want to learn about the history and cool facts while they are there. Therefore, scanning the QR codes.

1.1 Strategy

Our approach to achieving our strategy is to come up with 5 different interactive experiences that engage hikers, educate them, all while hopefully allowing and encouraging a connection in some form or another (with nature, oneself, family/friends, etc.)

*This approach is being taken because of our conversations with Cami and understanding her goals for the interactive trail experience.

2.0 Scope

Content to be developed

5 interactive Timpanogos Cave Trail experiences

- **Night Sky** - Educate hikers on the effects of light pollution as well as acknowledge that Timpanogos Cave is part of the Dark Sky Association.
- **Audio Experience** - Provide audio experiences of people from different native tribes to hear their stories.
- **Cave Multi-View** - This will show hikers the distanced view of the mountain and show them the three main spots as well as provide them with more information and images, Hansen Cave, Middle Cave and Timpanogos Cave.
- **Heads Up Game** - A game to entertain hikers as they wait to go into their cave tour. Familiarizes hikers with animals, plants, etc. around Timpanogos Cave.
    - **Word Scramble Game** - Heads Up was pivoted to word scramble game with outdoor and hiking words.
- **Flappy Bat** - This will be an entertaining way to educate hikers on the delicacy of the contents in Timpanogos Cave.
    - **Trail Birds** - Flappy Bat was pivoted to an audio trail birds experience where hikers can learn about the different birds they'll see and their calls they'll hear

2.1 Scope

Timpanogos Cave has a lot of content available which Cami can provide us with. However, the experiences we have chosen to do require us to gather a lot of new content. For example we are probably going to take our own image of the Night Sky experience, we need to go record audio experiences for the different tribes and the flappy bat game will be coded.

Content we do have available already is for the Cave Multi-View experience. We will need to collect information about each of the caves and images for them. We don't need it but we could possibly use an image of the cave zoomed out where you can see the three different cave entrances or where they are at on the mountain.

The great thing about the experiences we have chosen is they will not require content updates. These experiences will most likely not have any information that goes out of date so they can stay as is until someone wants to update them and change the content out.

These experiences will be loaded onto a webpage using code. Using pure code will work better with the raspberry pies and should provide hikers with the most ideal experience.

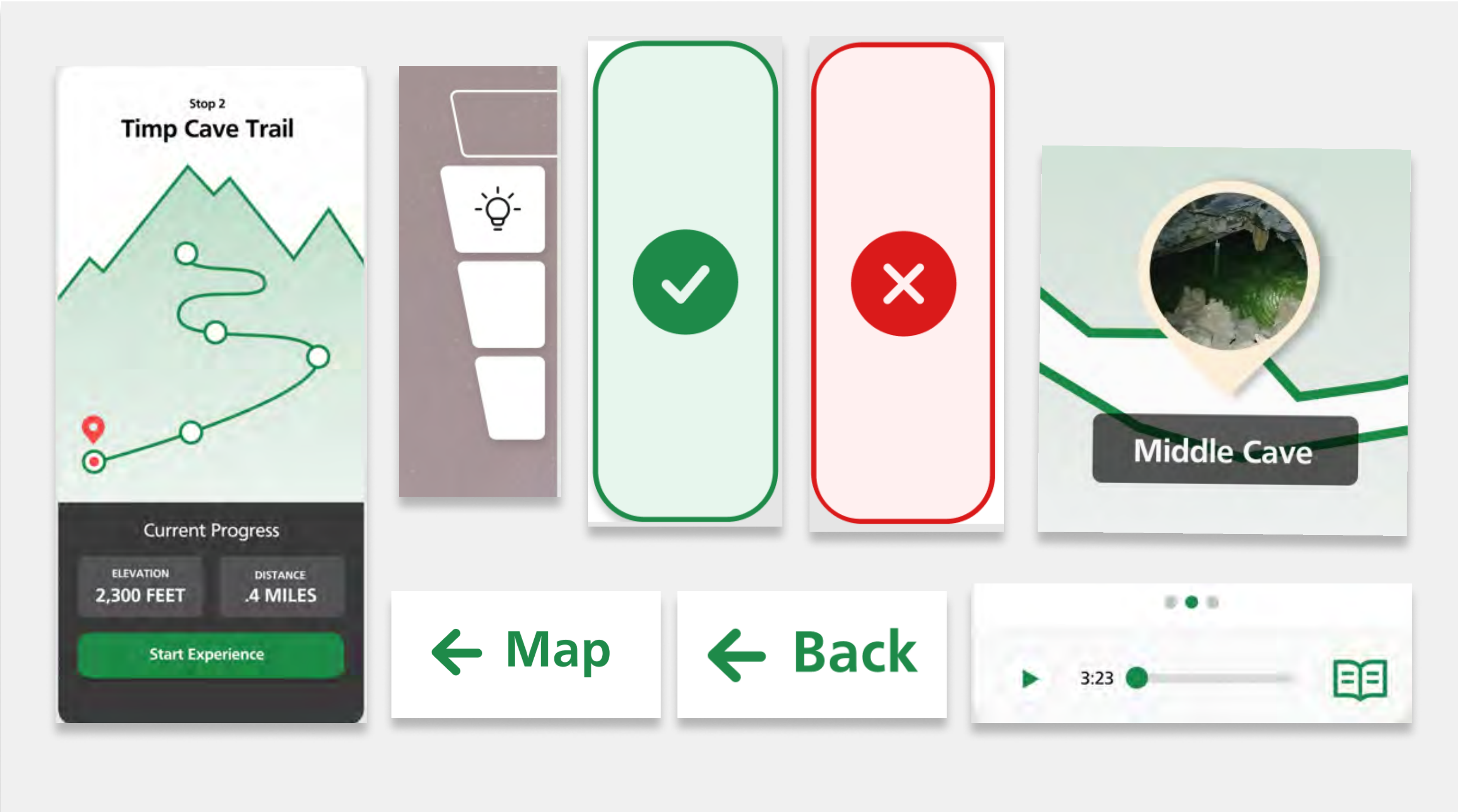
3.0 Structure

Our plan to best accommodate users is to provide really good, clear onboarding. We want to insert a sign towards the beginning of the trail that explains the gist of what the users will see and experience once they scan a QR code. This sign will also teach and explain to users how to scan and work a QR code if they are not already familiar.

A bonus to the experiences we have chosen is they are not connected to one another. Meaning, if a user skips or misses a QR code along the trail they will not be lost or confused. It will be its own experience.

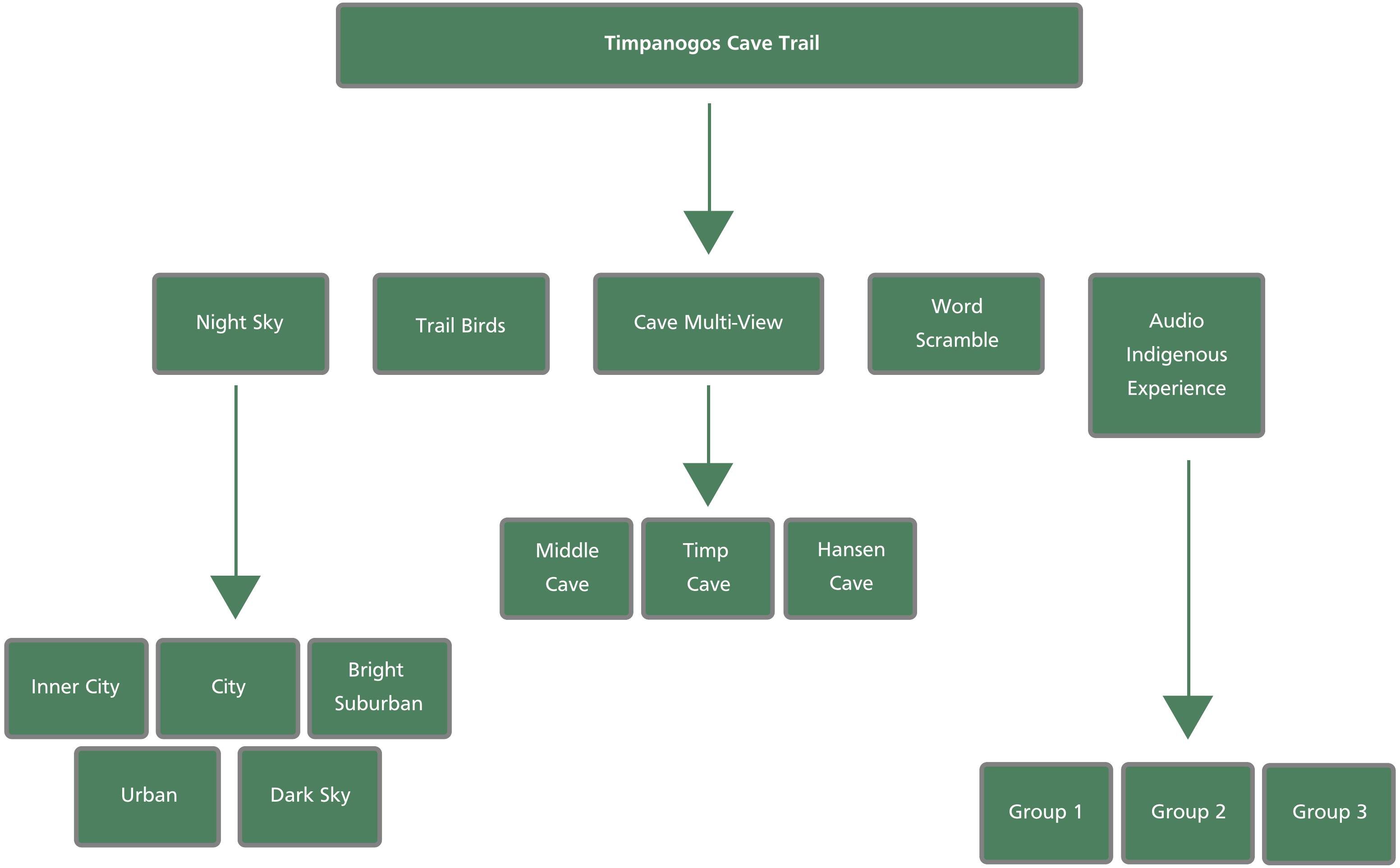
3.1 Structure

For navigation and labels we are using a few different tools. One of these is icons and labels and another is a distance map. Examples are displayed below.



3.2 Structure

Site Map of the QR code trail experiences.

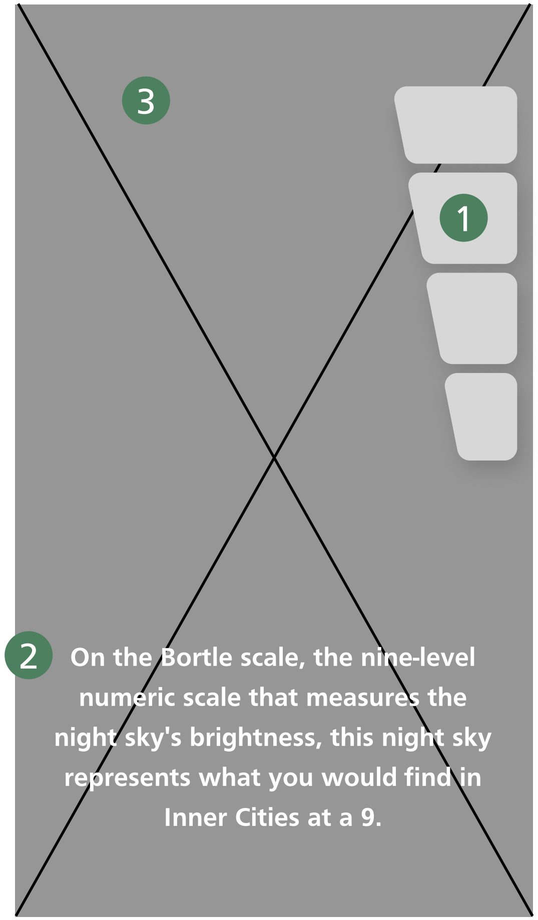


4.0 Skeleton

1 How to adjust and see different levels of light pollution

2 Light Pollution Night Sky description

3 Image



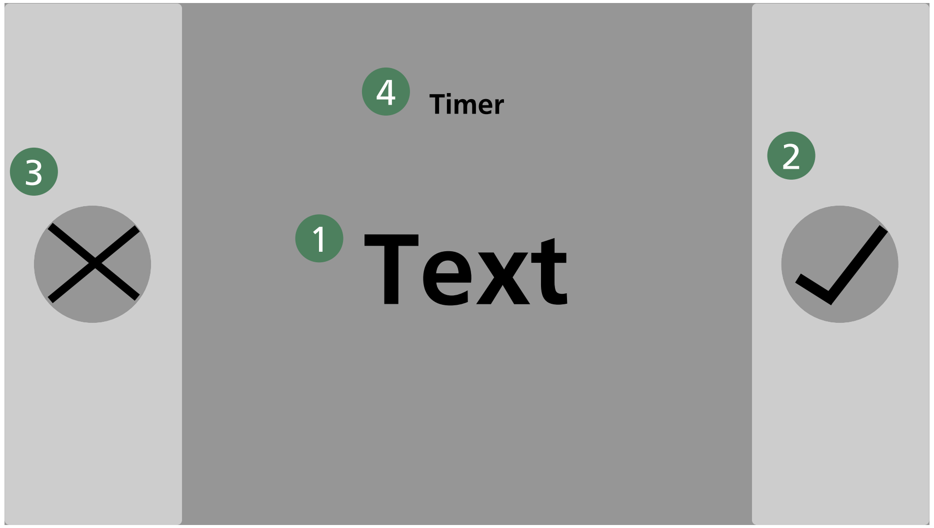
4.1 Skeleton

1 Rotating Word

2 If you guess the word right, tap check mark

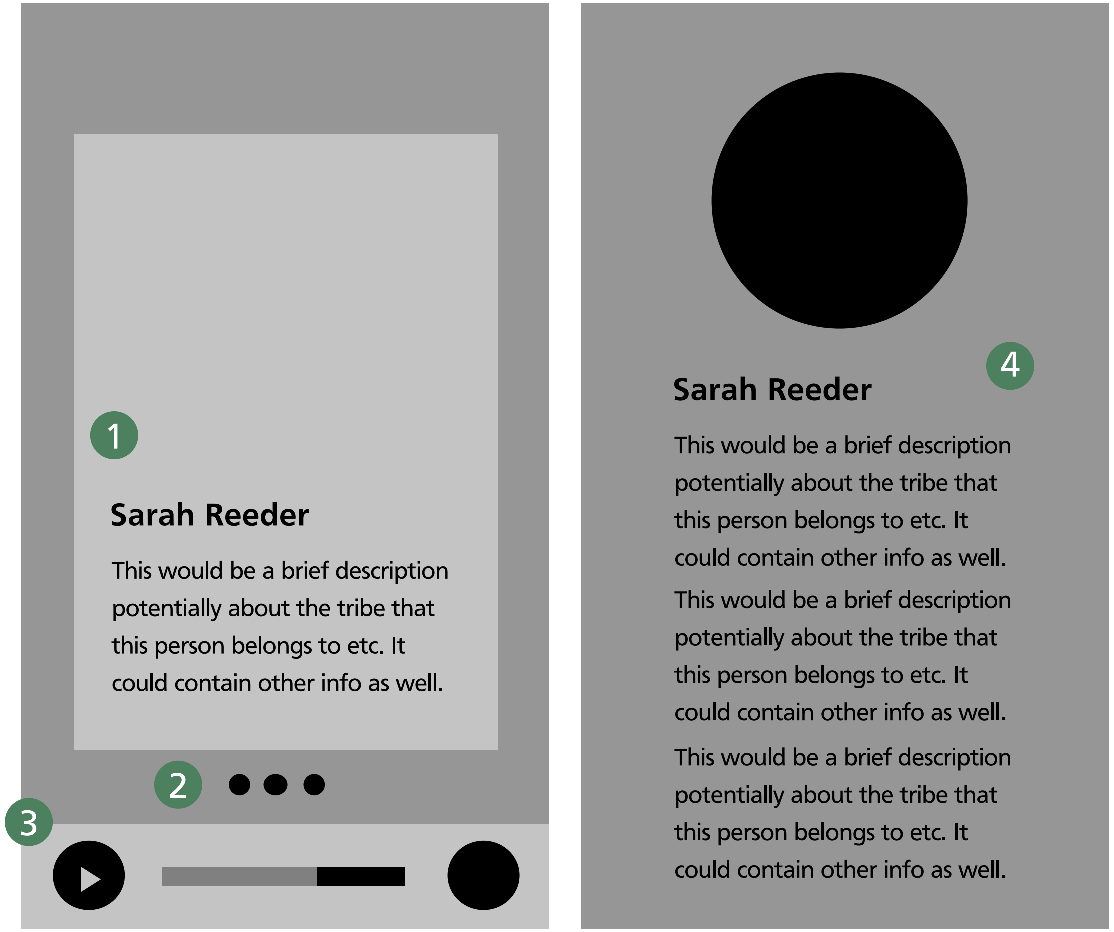
3 If you guess the word wrong, tap 'x'

4 Timer - see how many words you can guess right!



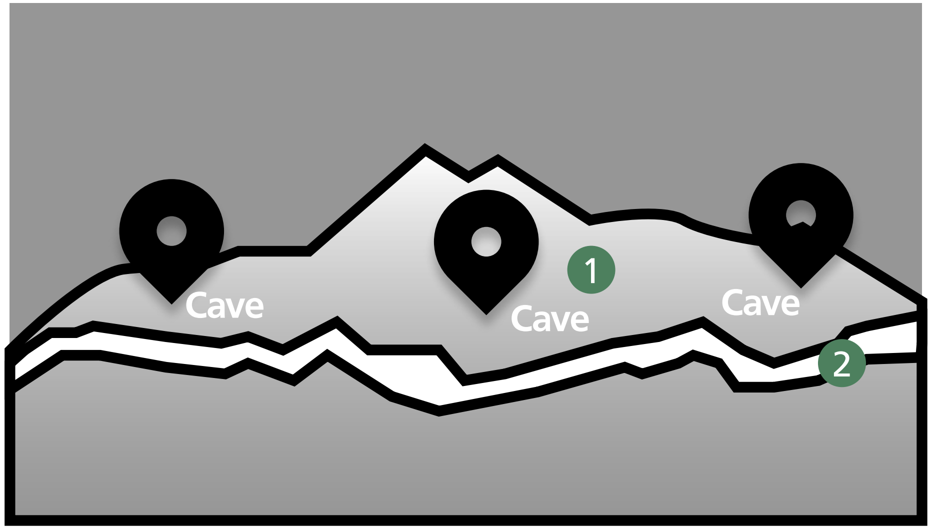
4.2 Skeleton

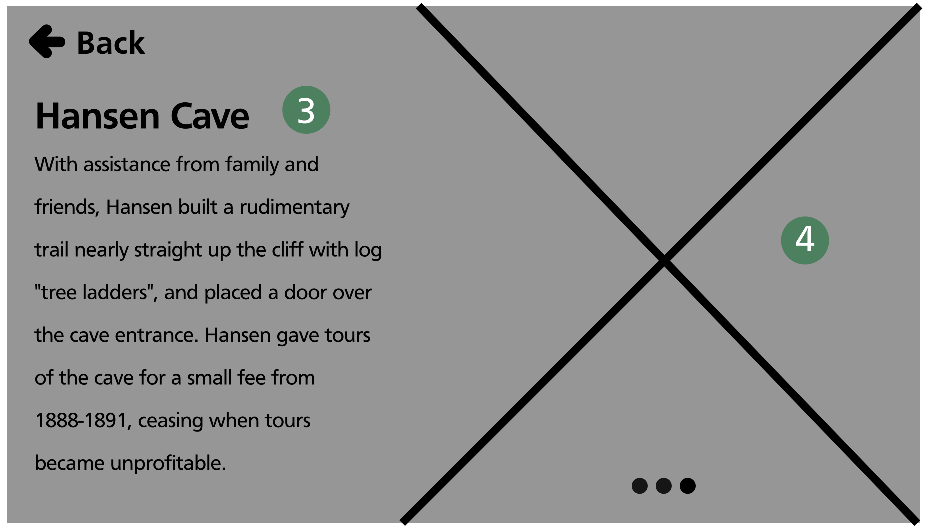
1. Image with name and description
2. Swipe through different native audio experiences
3. Play bar and text version
4. Interactive text experience



4.3 Skeleton

1. Cave name and location
2. Timpanogos Cave Trail
3. Cave name and history
4. Image gallery

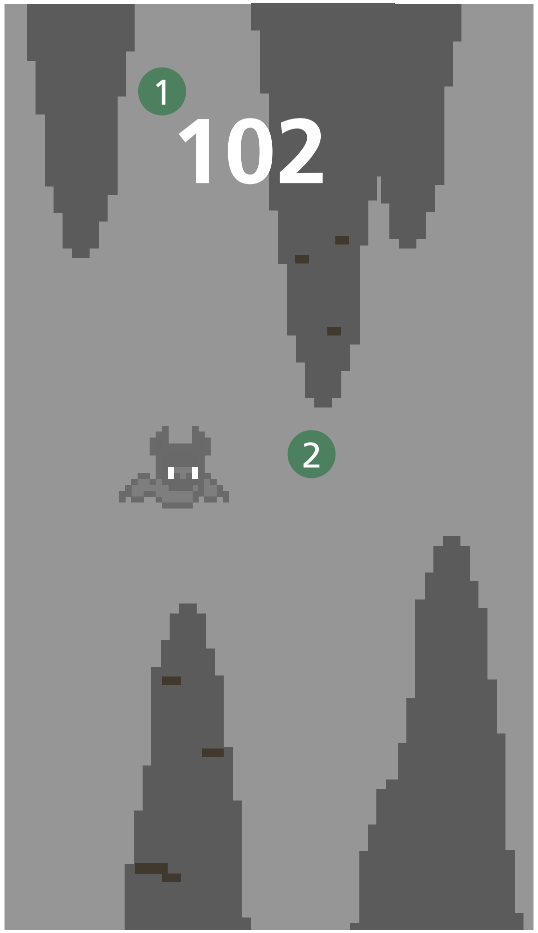




4.4 Skeleton

1. Flappy Bat score

2. Flappy Bat game - tap the bat and avoid the cave walls

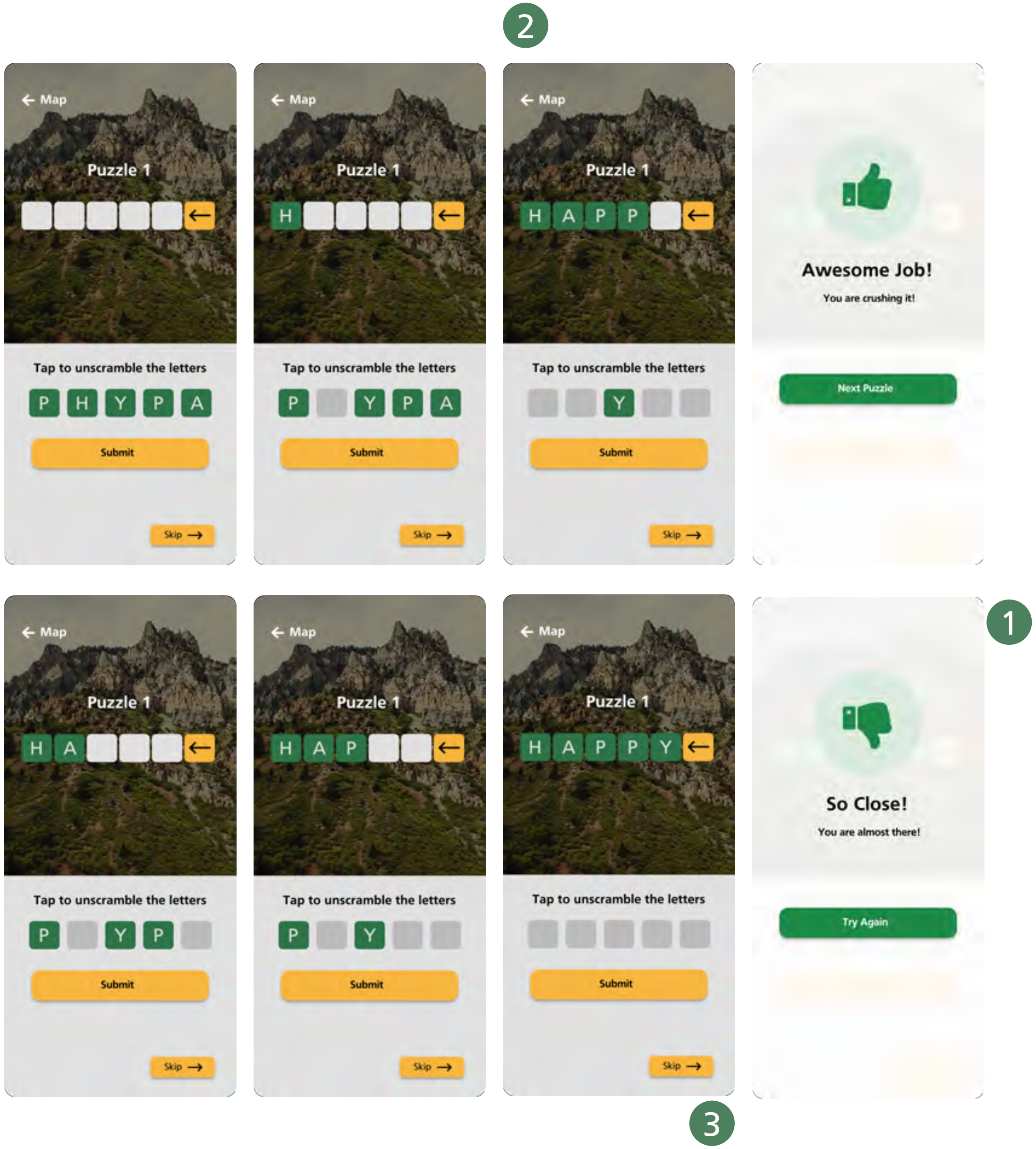


4.6 Hi-Fi Comps

1. Error and validation screens

2. Interactive buttons color coded for clairty

3. Skip button to pass on difficulty



4.5 Hi-Fi Comps

1. Persistent navigation - consistent thread throughout all experiences

2. Game over - educates on the delicate nature of cave structures

1

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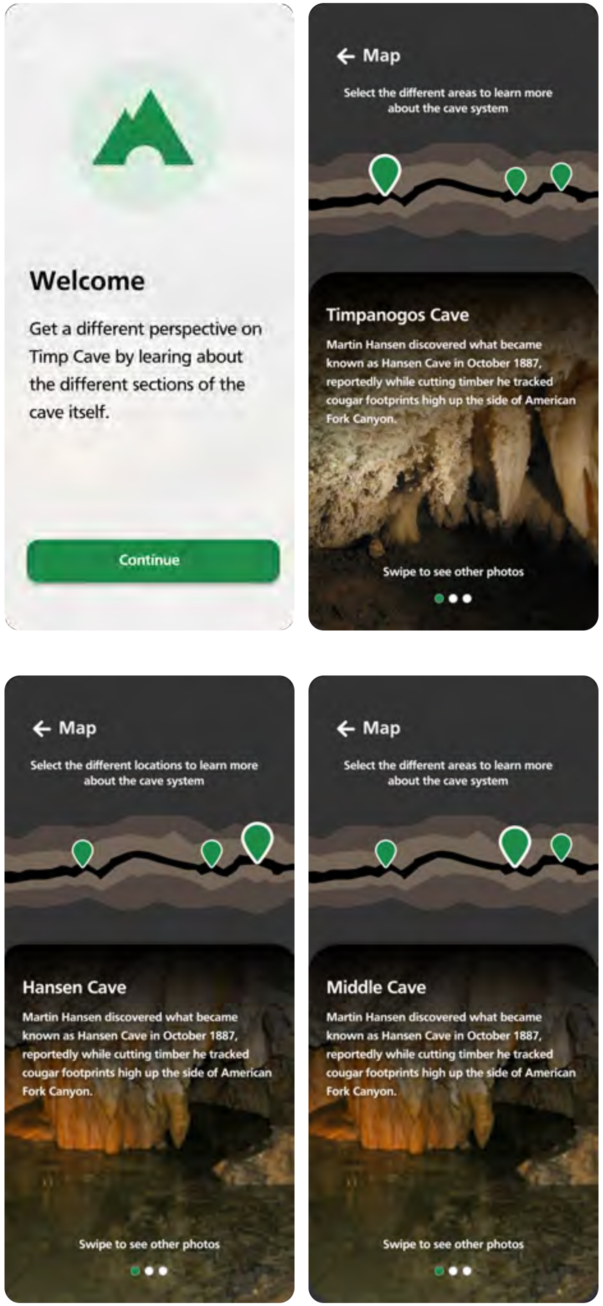
4.7 Hi-Fi Comps

1. Onboarding - preps user for the experience
2. Interactive icons for quick navigation
3. Pagination affordances for clear swipe call to action

1

2



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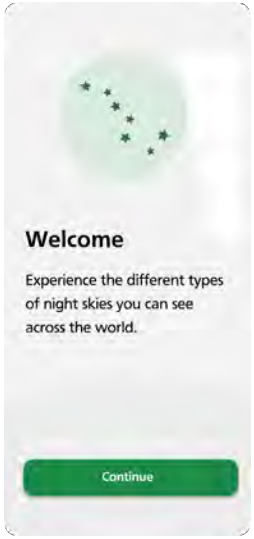
4.8 Hi-Fi Comps

1. Light toggle feature - matching design with increasing and decreasing levels of light pollution

2. Changing educational text

3. Night sky progression from clear, low level light pollution, and high level light pollution



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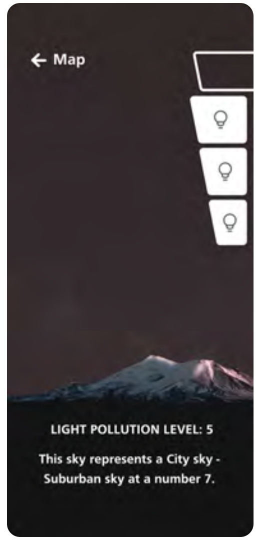

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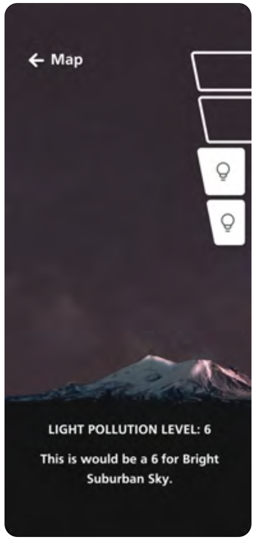
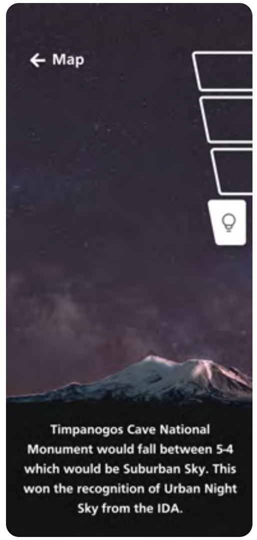
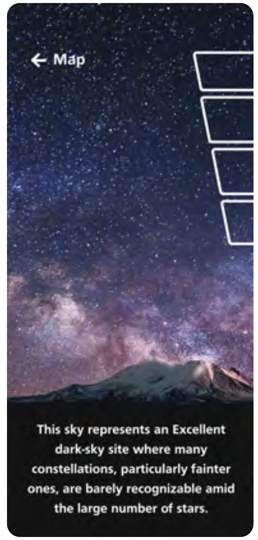






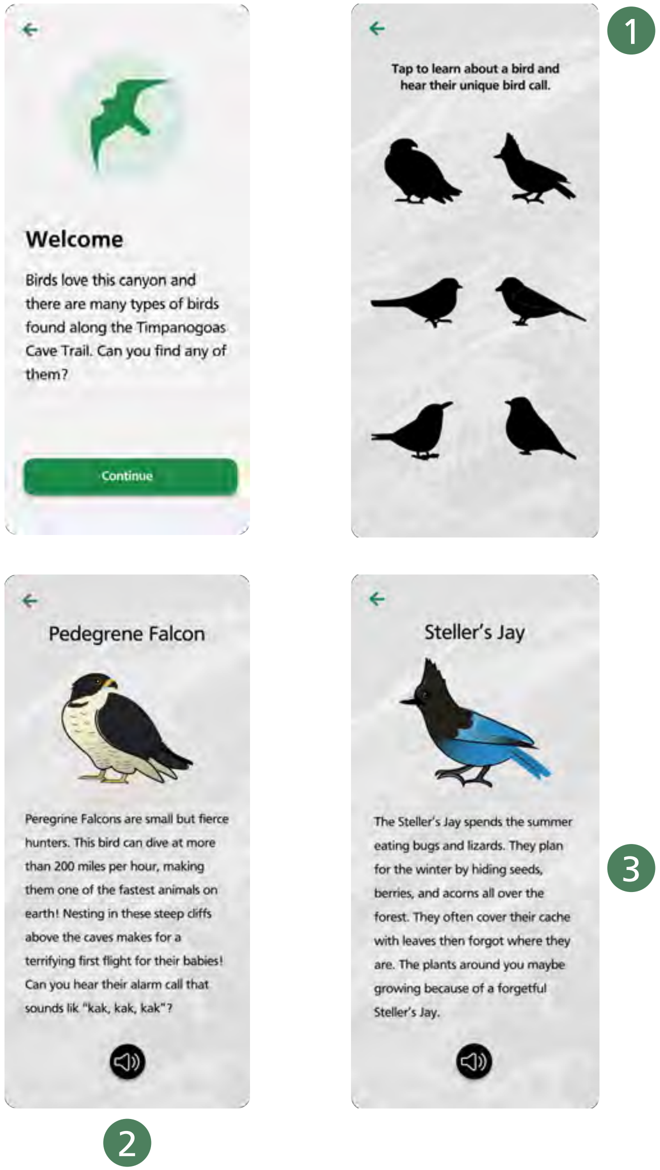
3

4.9 Hi-Fi Comps

1. Silhouettes of a few clickable birds. Tap on each to learn more

2. Audio button that plays that specific bird's call when tapped

3. Educational text about each bird



5.0 Surface

Color

For navigation and labels we are using a few different tools. One of these is icons and labels and another is a distance map. Examples are displayed below.

Primaries

Primary Green
#438750

Light Variant
#CCE5D1

Primary Tan
#FDF1DD

Red Accent
#F14C4C

Neutrals

White
#FFFFFF

Neutral 1
#F9FAFB

Neutral 2
#F1F3F6

Neutral 3
#3A3A3A

5.1 Surface

Typography

Our user base leans largely towards adults so we want to make sure that the type is large enough to read on mobile devices.

H1

Bold

Family	Size	Line Height
Frutiger	36px	36px

Lorem ipsum dolor

H2

Bold

Family	Size	Line Height
Frutiger	24px	32px

Lorem ipsum dolor sit amet,

Paragraph

Regular Bold

Family	Size	Line Height
Frutiger	20px	24px

Lorem ipsum dolor sit amet, conse

Technical Specifications

Equipment

Equipment

Solar Panels & Batteries

We worked with Dan Ostler to obtain larger solar panels and bigger batteries. The solar panels are able charge the battery even with only a few hours of sunlight. The batteries are able to last longer as well.

Raspberry Pi Hubs

The raspberry pi hubs contain the content. We are also able to track the # of users that visit each experience. However, we can only get the total counts at the end of the season and only after we take the hubs down since the usage stats are stored on the hubs.

Wi-fi Range

The range for the wi-fi at each experience is about 60-75 feet or so with minimal rock or tree cover. With an absolutely clear line of sight, approximately 300 feet.

Languages

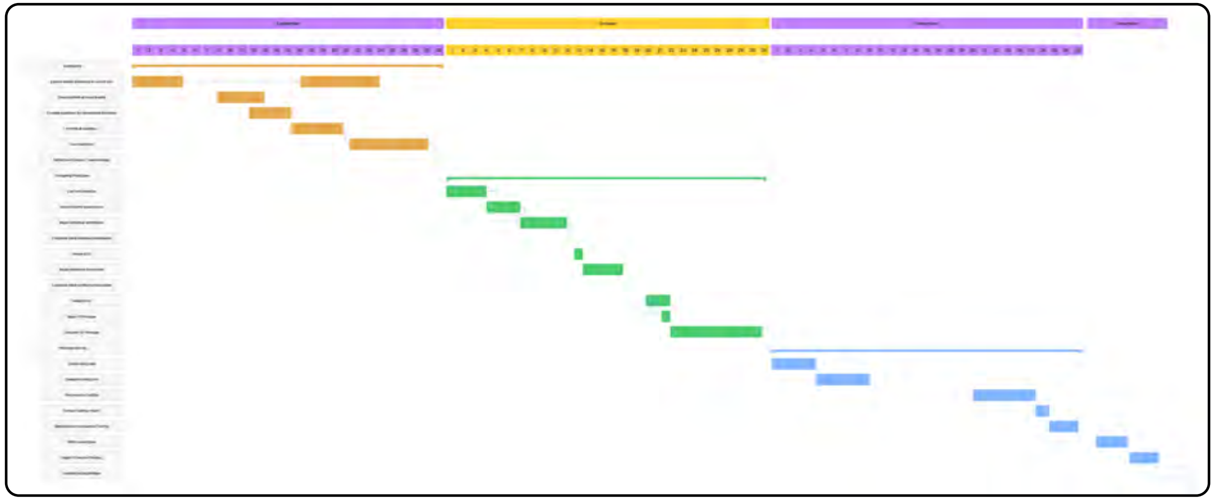
HTML, CSS, & JavaScript

We deicded it would be best to build the experiences using HTML, CSS, and JavaScript. These three languages together are light enough for the experiences to work well and load fast on the trail. Using any libraries or frameworks like React or Vue would have been clunky and/or slow on the raspberry pis.

Project Plan

6.0 Schedule

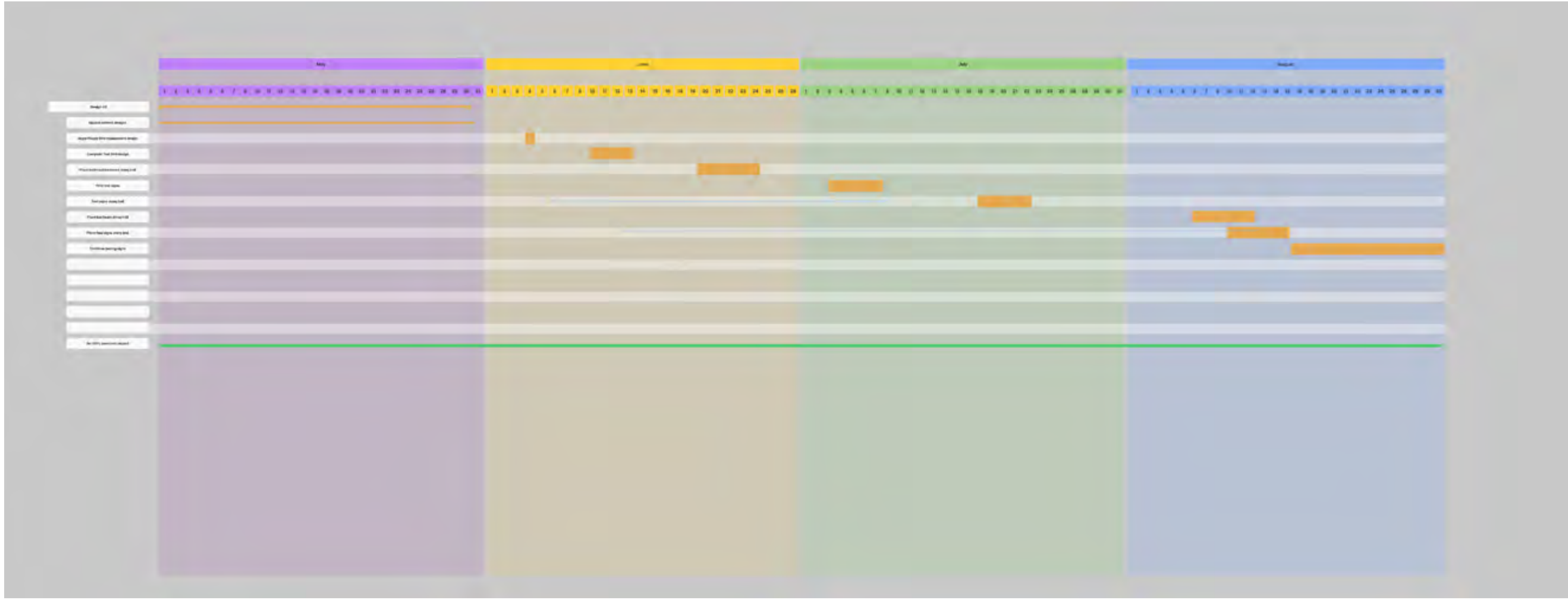
Fall 2021



Spring 2022

6.0 Schedule

Summer 2022



6.1 Development Process

Roles

Connor Chappell
Developer

Jake Carter
Designer

7.0 Budget

Task	Cost	Details	Total
Student Hours	$20/hr	~1500 hrs x 6 students	$30,000
Miles	$2,514.72	676 miles 4 trips	$2,514.72
Spark beacons	$400	5 x beacons	$2,000
Solar Panels	$456 x 4	Larger solar panels	$1,824
Spark Hub	$1,750	Visitor center hub	$1,750
Total			$38,088.72

8.0 Risk Assessment

Dependencies

- Not being able to get the audio assets and content for the Indigenous Experience from Cami when we would have liked
- Have to wait for Dan to bring new raspberry pi's and bigger solar panels to Timp Cave
- No work can be done on Timp Cave Trail through the Winter months while it is closed
- Have to wait for signs to be printed

Technical Risks/Contingencies

- Raspberry Pi's not having enough bandwidth
- Raspberry Pi's batteries lasting long enough
- Solar Panels not getting enough sunlight
- Making sure there is an internet connection at each QR code
- The content for each QR code needs to be pure HTML, CSS, and JS (no hype, React, etc.)
- Larger solar panels with a more robust battery system.

8.1 Risk Assessment

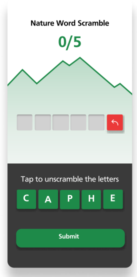
Change Control Process

We met frequently throughout the ideation process and clearly discussed this project would be a continued partnership to further build and develop new experiences. Any further development or changes to the scope would be added to the project backlog.

Appendices

Word Scramble Testing

Usability Testing - 12 respondents



Prototype

Respondents

Female - 6

Male - 6

Ages - 22 - 48

Prototypes tested 8

Results & Recommendations:

Users seem to have a preference for the Jerem prototype, because it makes the most sense, is the most intuitive, and allows for the best first interaction. Of all the experiences tested and means of navigation 7/12 users preferred the straight back arrow and how it clearly communicated the process of deleting selected letter. 4/12 users understood the selecting and tapping functionality for editing letter selection which clearly identified to our team that would no longer be pursued in future designs. Additional insights found were the added perceived complexity of a circular visual letter assortment. This design will also not be pursued in future iterations.

Appendices

Word Scramble Testing - Data

Female 24 all of them

Liked the straight arrow or keyboard style backspace button instead of the curved arrow. Wished there was a reset button and backspace.

Female 31 al of them Liked the straight arrow back button. Wished there was a reset button and a backspace button.

Male 32 all of them Liked the straight arrow and wished there was a reset and a back button

Female 23 all straight arrow makes it seem like a back button, curved arrow means reset. liked the colors, no preference on circle vs straight. arrow over back button. liked having the letters available over the keyboard.

Male 23 all likes straight line better than circle, likes straight back button better than curved , no button. Liked letter popping instead of down. No keyboard. likes 1st and 4th design.

Female 24 all She did not understand why on some of them the letters started at the top of the page. She liked how the red back was more clear.

Male 22 all Liked the straight arrow back button. Liked the red too but also thought maybe wish it was filled with an x to indicate delete. Liked the letters in a straight line better as opposed to a circle.

Female 46 all Liked when selecting the letters was at the bottom. Felt more natural like a keyboard. Didn't like the keyboard however because it presented too many letter options that weren't necessary. Especially when letters are already shown. Wanted reset button

Male 48 all Would have liked the backspace/delete button to have an x inside. Wished there was a reset button. Preferred straight to circle. Liked the mountain background.

Appendices

Word Scramble Testing - Data

M 31 all

First experience was clear. Immediately got that the arrow was a simple back button. Asked why, “because it’s all a single file leading up to the red arrow, it just seemed like that’s what it would do”. Color is nice and bright, overall design was boring. Didn’t notice the type overhead. functionality wasn’t clear, didn’t like the tapping of the prototype took 4 clicks between screens to full understand functionality. Liked the color / overall design thought it was inviting and bright. Liked the arrangement of the letters for this version better than a singular line even thought the clarity of the functionality was better.
"Liked the color immediately. Asked why this was such a departure from the others. Immediately saw the word. Didn’t seem to notice he took his phone in both hands (slowly), didn’t complain so probably not a big enough hinderance. Didn’t see the timer

Didn’t like the color gradient, thought it felt dated. Noticed the timer, asked how it would interact asked how he imaged it working. It would shut him out of the puzzle and have him start over (almost like that popup game from a million years ago and have to start again). Commented on functionality being higher up.

Very wordle but in a good way. Didn’t notice the letters above right off, trying entering some random letters before realizing it was still the directive of a word scrambler. Likes this idea of more puzzles, like wordle has him coming back everyday to the game. Didn’t mention anything about functionality at the bottom (not comment probably a good thing, its just intuitive and goes under the radar)

Understood from the previous brief. Asked what the count up top was for, he imagined it was to adjust the words and add bluff letters (I did mention the possibility of harder levels was a concept we discussed). Liked the thumbs up, felt like the closing screen in wordle which made him ask if we’d be able to share it (asked if that was important to him) he said not really but it’d be cool from a functionality perspective.

Appendices

Word Scramble Testing - Data

M 25 all
Started going the other way. Asked about the map liked the inclusion because he likes that kind of stuff (asked if he liked hiking) eh, not his favorite but he does it but likes when there's stuff to interact with and informs him of stuff. Liked the animation of the letters flying up, kind of slow animation though. Asked where the timer was from the previous screen. Thought that would be cooler (having a timer).

Wordle cool. "Lets see, g" didn't realize the letters where not the name of the game but letters he could use felt like they were just up there.

Felt top heavy and commented on the colors being off from previous ones (if this is the game on top of the hill would it dislodge them if it didn't fit the overall experience assuming they've entered all the experiences?) "oh boy its on" reaction to the timer. Did have to take the phone in both hands.

Liked the colors. "Do I have to tap those color buttons.." Mentioned it just felt like more work even though the last one was the same except for in a straight line. Shrugged it off.

Paused for a min. Liked that it was at the bottom but there really wasn't much to tell him what to do. Instinctively went to tap the letters one by one to remove.

Thought the letters where really high "making me work for those letters" (probably bridging the gap of the letters and the spaces to be unscrambled.) Understood what the back space was for and how it functioned.

Appendices

Word Scramble Testing - Data

F 27 all
Didn’t like that the letter were too high, if users were going to be playing on the phone. Had to move position of her hands but that might just be suzes tiny hands.

Did mention how it felt unfinished but liked the appeal of the curved letter “that’s fun”. “The outline could be in the shapes of stuff which could be cool”.

Liked the introduction of colors “feels like one of the jackbox games” Wondered if there were going to be any haptics or touch response I said probably not. Did not mention or see the timer really (this may just be because it’s not moving time.)

Saw the timer first. Thought the space at the bottom was larger and it made her think of tapping the letters with her tiny hands. Would be stressed having a time go. “Can I come back to this one if I don’t get it in time?” Under the impression that she’d be prompted to move to the next one and it was a scoring as high as you can or she said another way it could be done like that instagram filter where the timer continually resets itself after each right answer.

Wordle mentioned again. She did see the letters up top liked that the keyboard was easy access. Thought it was pretty straight forward and liked the retro feeling of the design?

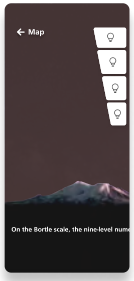
Liked this entry point of the map and onboarding. Wondered if there would be words that would not use all the letters. It made sense to count letters but it could also be used to track on which puzzle she’s in. If there is more or if we have a limited amount. She liked the “tossing” animation of the letters moving into and out of place.

Appendices

Night Sky Testing

Usability Testing - 8 respondents

Respondents
Female - 6
Male - 2
Ages - 16 - 60
Prototypes tested 1



Prototype

Results & Recommendations:
Users were asked to test a second iteration of the Night sky experience and specifically whether the light adjusting functionality was intuitive and easy to understand. 6/8 users found no problem in navigating the toggle light adjustment. 2/8 users specifically commented on the content the experience was trying to convey as interesting and came away having learned something. 3/8 respondents commented on copy being confusing and or felt that the experience did not do a good job at correlating the data with the interactive experience of adjusting the brightness. From this we are going to be solidifying a more consistent structure to the copy to better convey to the users the correlation of adjusting light pollution and information provided.

Appendices

Night Sky Testing - Data

P1 16 She thought the night sky experience with the descriptions on the bottom was better and made more sense. With the 'i' didn't realize we were doing light pollution

P2 54 Experience with descriptions on the bottom was way less confusing. Didn't connect that the different tabs correlated to different levels of light pollution. Users wonder what's next when they are done.. is there a way for us to move them forward

P3 53 i' means you need help, i didn't need help lol. Not self explanatory to click the white bars. Second one is better with better explanation. Add a # to the button saying I want to see a 9 night sky etc.

P4 35 Tapped on red then green, went about tapping the toggle before reading, noticed they changed at first thinking those were directions.

P5 31 First screen on first read had three buttons, transitions came across a laggy play with speed, thought it was fun idea, drawn to touch the white bars.

P6 60 Clicked all the squares on first screen. Night sky was simple enough, text size seemed to be a problem having to adjust how close she was to the phone was able to tap light toggle fine.

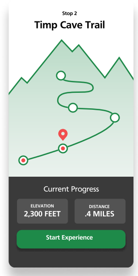
P7 27 Liked, maybe say welcome to start of expereince, easy to navigate, liked the toggle a lot.

P8 24 Everything went smoothly. They know how to interact with the interface and didn't have any questions.

Appendices

Audio Experience

Usability Testing - 8 respondents



Prototype

Respondents

Female - 6

Male - 2

Ages - 16 - 60

Prototypes tested 1

Results & Recommendations:

Users were asked to test a third iteration of the Native Audio experience which would allow users to listen to stories from the perspective of Native persons and their relationship with Timp cave and the surrounding lands. 4/8 users did not readily notice the swipe-able navigation of the experience which would allow users swipe between different stories. 2/8 users misunderstood our way-finding iconography as bios about the different native story tellers rather than an transcript of the provided audio. 1/8 users commented on the complexity of this experience and how it would be difficult if onboarding was simply skipped through. 2/8 users did mention this experience based on the content was their favorite experience they have tested. Recommendations are to increase the visibility of the pagination and perhaps increasing affordances that allow user to notice the navigation structure and perhaps looking at different way-finding icons that better represent transcript.

Appendices

Audio Experience Testing - Data

P1 16 She thought by clicking the book you were going to learn more about the person - not that the text and audio were the same thing. She didn't read any of the description pages.

P2 54 Didn't realize you could swipe to different people, but said she should've realized cause that's what the dots mean lol. She knew what the book meant though. If you click past the description page quick, you could get confused what the experience was.

P3 53 This one didn't have issues. For the map page though we could change to how far they are to the cave instead or make the first checkpoint not look like the beginning since the first QR code won't be at the beginning of the trail

P4 35 Like the best, didn't notice the scroll at first just tapped, book was also commented on - was it their bio?

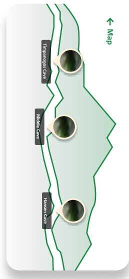
P5 31 Could map be more than just back? Liked the scroll, went through without asking commented aftwards, also thought the book was bio.

P6 60 Did not notice scroll, eventually found it. Liked the size of text.
P7 27 Thought it was a pretty landing page, no problems, commented the difficulty may be because of the white in the next photo.

Appendices

Cave Multi View

Usability Testing - 8 respondents



Prototype

Respondents

Female - 7

Male - 1

Ages - 16 - 60

Prototypes tested 1

Results & Recommendations:

Users were asked to test the Cave multi view experience which allows users to view the cave system through an X-ray cut out perspective. For this iteration we had a forced landscape view which required users to turn their phone to interact with the cave system. 7/8 users registered the forced perspective and reoriented their phones accordingly. 4/8 users carried out this reorientation with no comment rather a simple understanding that this experience was in a forced perspective. 6/8 users recognized that once you clicked within a specific cave you would be able to navigate through photo galleries of the different systems. Recommendations are to increase the definition to users who were not able to readily see cave galleries by increasing contrast in way finding / navigation and adding dialogue in onboarding that mentions these galleries so they actively seek them out when going through the experience.

Appendices

Cave Multi View Testing - Data

P1 16 Didn't notice when you clicked on the different caves that the images were galleries

P2 54 Noticed the galleries.

P3 53 Make it so you can get to the other caves without going back to main page.

P4 35 Noticed galleries, turned phone just fine

P5 31 Gradient mentioned, Registered had to move phone sideways no comment, text would seem difficult to read for other users on landing screen, noticed galleries.

P6 60 Didn't register that phone needed to be turned sideways till she clicked the bubble (probably outlier), did not notice galleries.

P7 27 Turned phone just fine, noticed galleries, mentioned bubbles blocking the cave view.

P8 24 Turned phone just fine. Didn't have any problems.

Appendices

Heads up

Usability Testing - 7 respondents

Respondents

Female - 7

Male - 1

Ages - 16 - 60

Prototypes tested 1

Prototype

Results & Recommendations:

Users were asked to test the Cave multi view experience which allows users to view the cave system through an X-ray cut out perspective. For this iteration we had a forced landscape view which required users to turn their phone to interact with the cave system. 7/8 users registered the forced perspective and reoriented their phones accordingly. 4/8 users carried out this reorientation with no comment rather a simple understanding that this experience was in a forced perspective. 6/8 users recognized that once you clicked within a specific cave you would be able to navigate through photo galleries of the different systems. Recommendations consider other means of playing the game from an accessibility perspective for those who have fine motor difficulties. Being cognizant of words in the experience and selecting those that would be recognizable to the broadest set of users.

Appendices

Heads up View Testing - Data

P1 16 She didn't know what stalagtite was – do we want to be careful with our terms and verbiage we use?

P2 54 She didn't have an issue with this one!

P3 53 Went good, no issue

P4 35 Blurred game made her think of next steps before reading text, the blur distracted her action

P5 31 Blur on game was commented on, can text not overlap? maybe bolded like last screen? buttons overlapped slightly

P6 60 Liked the game, went through the motions saying it'd be easy to play might drop phone. Screen super clear on direction

P7 27 A lot of text in beginning maybe through illustration direction clear overall nav.

Appendices

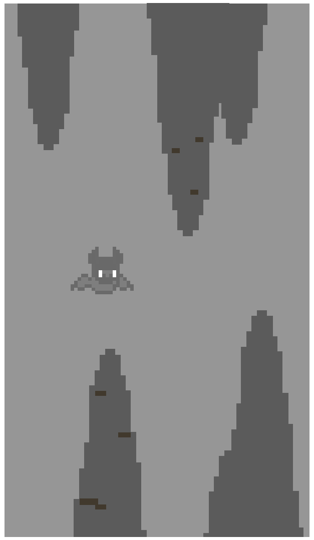
Flappy Bat Pivot

Our original plan was to have one of our experiences be a game called, “Flappy Bat.”

After testing and developing it, we realized how much harder it was to build than we previously thought.

We then decided to pivot and create a new experience. That is when we came up with the Trail Bird experience.

A previous group designed a very similar experience about the bird along the trail that they also did testing for. We just implemented their idea and made our own experience that matched our design aesthetic.



Appendices

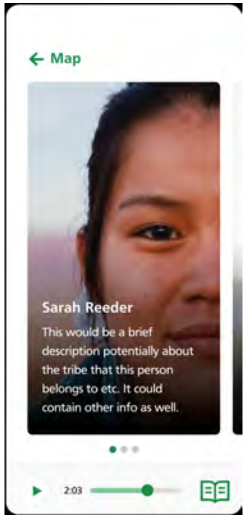
Indigenous Experience Pivot

Our original plan was to have an experience that focused on the indigenous people that lived in the valley below American Fork Canyon.

The experience was going to consist of stories told by members of a tribe in Utah and people would be able to listen to the stories while they hiked the trail.

We learned throughout this project that it was difficult to stay in communication with the tribe leaders and members. So in the end we were not able to create that experience.

We still wanted an Indigenous experience though, so we created a very simple one that users can scroll through and read about the indigenous people who who used to live in the valley.



Appendices

Signage

Usability Testing - 7 respondents

Prototype

Respondents

Female - 5

Male - 2

Ages - 24 - 54

Prototypes tested 3

Results & Recommendations:

Users were asked to test the Cave multi view experience which allows users to view the cave system through an X-ray cut out perspective. For this iteration we had a forced landscape view which required users to turn their phone to interact with the cave system. 7/8 users registered the forced perspective and reoriented their phones accordingly. 4/8 users carried out this reorientation with no comment rather a simple understanding that this experience was in a forced perspective. 6/8 users recognized that once you clicked within a specific cave you would be able to navigate through photo galleries of the different systems. Recommendations consider other means of playing the game from an accessibility perspective for those who have fine motor difficulties. Being cognizant of words in the experience and selecting those that would be recognizable to the broadest set of users.

Appendices

Signage Testing - Data

Sign 1

Female college Noticed the 1/5 but thought it was irrelevent to the test, liked the sign but a lot to look at, knew the idea was to scan the code.

Female college Didn't notice the 1/5 on the sign, a lot to look at, would scan qr code.

Female college made the most sense to her, this sign caused her to read more of the small print, noticed the wifi scan.

Males college thought this sign made the most sense, thought the numbers represented light pollution levels or dates.

Females college this sign was their favorite, noticed the 1-5 markers and thought there would be more signs, said it was aesthetically pleasing.

Female 54 Easiest to understand, wanted the 1-5 a little bigger, knew to scan qr code.

Male 54 thinks adding "stop" to the numbers will be helpful, doesn't think you need the "once" in the wifi sentence, would scan big qr code.

Sign 2

Females college thought this sign was confusing, they didn't understnad the tab affect at first. confused why there was two qr codes.

Females college confused when she first saw it. Not intrigued to scan anything.

Males college Couldn't tell what the background picture was. thought the numbers represented the light pollution levels.

Females college would scan qr code, but confused by the tabs.

Female 54 Knew to scan qr code, knew what the tabs meant, liked design

Male 54 didn't understand the tabs at first, thinks you need to add "stop 1", would scan little qr code if got an error on the big one. liked this one the best!

Appendices

Signage Testing - Data

Sign 3

Females college this sign made the most sense to them.

Female college would scan qr code, didn't understand what the "1" was for.

Males college thought the "1" was a date or that you were on level "1", knew to scan qr code

Females college thought they were at step 1, would scan qr code.

Female 54 knew to scan qr code, thought you were at step 1.

Male 54 thought it was confusing but would scan qr code.

Appendices

Next Steps

- Create an onboarding experience at the beginning that will tell hikers about the experiences that they will encounter while they hike.
  - Possibly a home page that just explains what the experiences are and the steps to access them.

- Check the QR codes on the signage to make sure there has been no tampering.

- Create a new indegenous experience that is more interactive.

- Coordinate with Dan Ostler about getting the user visits / hub stats for each experience. The hubs have to be taken down at the end of the season in order to retrieve that data.

- Continue testing each experience to make sure wifi connection is good and that the experiences are workig properly.

Appendices

Google Drive Link

Below is a link to all the deilverables for the project.

https://drive.google.com/drive/folders/1XqSiL5KRL93M-DGOWk4uqKga5xuvU_xe?usp=sharing

Appendices

Photos - Installation

Appendices

Photos - Signs

Appendices

Time Logs - Conner C.

Week 1	Description	Time In	Time Out	Total Hours
5/18/2022	Re coding cave cutout experience	6:00:00 PM	10:00:00 PM	4:00
5/24/2022	Taking down old signs on trail	5:00:00 PM	7:45:00 PM	2:45
5/25/2022	Meeting with instructor	4:00:00 PM	4:30:00 PM	0:30
5/25/2022	Meeting with instructor assignment	7:00:00 PM	7:30:00 PM	0:30
5/26/2022	Fixing Flappy Bat assets for possible redo/re coding	6:00:00 PM	7:30:00 PM	1:30
5/26/2022	Flappy Bat coding research for possible redo/ re coding	8:00:00 PM	8:45:00 PM	0:45
Conner Chappell				**10:00**
Week 2	**Description**	**Time In**	**Time Out**	**Total Hours**
6/1/2022	Team meeting	4:00:00 PM	4:30:00 PM	0:30
6/1/2022	Home/landing/onboarding page design	6:00:00 PM	10:00:00 PM	5:00
6/2/2002	Home/landing/onboarding page design	6:00:00 PM	10:00:00 PM	4:30
Conner Chappell				**10:00**
Week 3	**Description**	**Time In**	**Time Out**	**Total Hours**
6/6/2022	Home/landing/onboarding page design	7:00:00 PM	9:00:00 PM	2:00
6/7/2022	Home/landing/onboarding page design	7:00:00 PM	9:00:00 PM	2:00
6/8/2022	Home/landing/onboarding page design	7:00:00 PM	9:00:00 PM	2:00
6/9/2022	Home/landing/onboarding page design	7:00:00 PM	9:00:00 PM	2:00
Conner Chappell				**8:00**
Week 4	**Description**	**Time In**	**Time Out**	**Total Hours**
6/13/2022	Flappy Bat coding research for possible redo/ re coding	6:00 PM	10:00 PM	4:00
6/14/2022	Flappy Bat coding research for possible redo/ re coding	6:00 PM	10:00 PM	3:00
6/15/2022	Team Meeting	4:00 PM	4:30 PM	0:30
6/16/2022	Experimenting with navigation between experiences	6:00 PM	8:00 PM	2:00
6/17/2022	Technical meeting with Dan Ostler	5:30 PM	6:00 PM	0:30
Conner Chappell				**10:00**
Week 5	**Description**	**Time In**	**Time Out**	**Total Hours**
6/21/2022	Drove up to Timp Cave to put up raspberry pis	8:30 AM	11:30 AM	3:00
6/22/2022	Meeting with Instructor	4:00 PM	4:30 PM	0:30
6/22/2022	Team meeting	4:30 PM	5:00 PM	0:30
6/22/2022	Bird experience research	6:00 PM	8:00 PM	2:00
6/22/2022	Bird experience design	8:00:00 PM	10:00 PM	2:00
6/23/2022	Coding home page tests	6:00 PM	8:00 PM	2:00
Conner Chappell				**10:00**
Week 6	**Description**	**Time In**	**Time Out**	**Total Hours**
6/28/2022	Bird sounds research	6:00 PM	7:00 PM	1:00
6/29/2022	Team meeting	4:00 PM	4:30 PM	0:30
6/29/2022	Polishing 4th of July test survey	7:30 PM	8:00 PM	0:30
6/30/2022	Installing raspberry pis on trail	8:30 AM	5:30 PM	9:00
Conner Chappell				**11:00**
Week 7	**Description**	**Time In**	**Time Out**	**Total Hours**
7/5/2022	Drove up to test dark sky experience	5:00 PM	6:45 PM	1:45
7/5/2022	Bird research and going through previous group's bird assets	7:00 PM	10:00 PM	3:00
7/6/2022	Meeting with instructor	4:00 PM	4:30 PM	0:30
7/7/2022	Drove up to test dark sky experience	5:00 PM	6:45 PM	1:45
7/8/2022	Drove up to test dark sky experience	5:00 PM	6:45 PM	1:45
Conner Chappell				**8:45**
Week 8	**Description**	**Time In**	**Time Out**	**Total Hours**
7/10/2022	Drove up to test dark sky experience	1:00 PM	2:30 PM	1:30
7/11/2022	Drove up to test dark sky experience	5:00 PM	6:30 PM	1:30
7/12/2022	Drove up to test dark sky experience	5:00 PM	6:30 PM	1:30
7/13/2022	Team meeting	4:00 PM	4:30 PM	0:30
7/13/2022	Coding bird experience	6:00 PM	8:30 PM	2:30
7/14/2022	Coding bird experience	6:00 PM	8:30 PM	2:30
Conner Chappell				**10:00**
Week 9	**Description**	**Time In**	**Time Out**	**Total Hours**
7/17/2022	Coding bird experience	7:00 PM	10:30 PM	3:30
7/18/2022	Drove up to test network connections	8:30 AM	11:30 AM	3:00
7/18/2022	Coding bird experience	7:15 PM	10:45	3:30
7/20/2022	Meeting with instructor	4:00 PM	4:30 PM	0:30

Appendices

Time Logs - Conner C.

7/20/2022	Team meeting	4:30 PM	5:00 PM	0:30
Conner Chappell				**11:00**
Week 10	**Description**	**Time In**	**Time Out**	**Total Hours**
7/21/2022	Meeting with Dan Ostler about updating content	3:30 PM	4:30 PM	1:00
7/24/2022	Attempt at updating content on the trail / testing experiences	3:00 PM	5:30 PM	2:30
7/25/2022	Meeting with Dan Ostler about updating content	3:30 PM	4:00 PM	1:00
7/27/2022	Team meeting	4:00 PM	4:30 PM	0:30
7/30/2022	Attempt at updating content on the trail / testing experiences	8:00 AM	12:30 AM	4:30
Conner Chappell				**9:30**
Week 11	**Description**	**Time In**	**Time Out**	**Total Hours**
7/31/2022	Documented updating content steps	1:30 PM	2:00 PM	0:30
8/3/2022	Meeting with instructor	4:00 PM	4:30 PM	0:30
8/3/2022	Team meeting	4:30 PM	5:00 PM	0:30
8/3/2022	Worked on design document	5:30 PM	8:30 PM	3:00
8/4/2022	Worked on design document	5:30 PM	8:30 PM	3:00
8/5/2022	Worked on design document	7:00 AM	9:00 AM	2:00
Conner Chappell				**9:30**
Week 12	**Description**	**Time In**	**Time Out**	**Total Hours**
8/7/2022	Final Refleciton and team evaluation	4:00 PM	5:30 PM	1:30
8/8/2022	Worked on design document	4:00 PM	5:00 PM	1:00
8/9/2022	Worked on design document	4:00 PM	5:00 PM	1:00
8/10/2022	Putting up signs on the trail / updating content	8:00 AM	1:00 PM	5:00
8/11/2022	Worked on design document	6:00 PM	8:00 PM	2:00
Conner Chappell				**10:30**

Appendices

Time Logs - Jake C.

Jake Carter

Date	Description	Time In	Time Out	Hours
Week 1				
1/11/2022	Research on the Beit Lehi site in Israel and worked did UVU WDD Student Career Survey	9:00	10:30	1.5
1/12/2022	Researched Beit Lehi	7:00	8:00	1
1/13/2022	Did more research on Beit Lehi archeological site	5:00	6:00	1
1/14/2022	Worked on Creative Brief, looked over project initiation form and more research and reviewed course deliverables	10:30	1:00	2.5
			Total:	6
Week 2				
1/20/2022	Watched recording of first team meeting. Met with Dan, Emily, and the team to figure out next steps and a timeline	4:30	6:00	2.5
1/21/2022	Researched ideas for a new game for an experience	6:00	7:00	1
1/22/2022	Reasearched for new game ideas and worked on designing new game	11:30	5:00	5.5
1/22/2022	Sketched Ideas for game	5:30	6:30	1
			Total:	10
Week 3				
1/25/2022	Designed game prototype for new experience	10:00	11:00	1
1/25/2022	Team Meeting and worked on prototype for game experience	5:00	6:30	1.5
1/27/2022	Worked on word scramble prototype and looked over other experiences	11:30	2:00	2.5
1/27/2022	Worked on new game idea and prototype	4:00	5:00	1
1/27/2022	Team Meeting and worked on prototype	5:00	6:30	1.5
1/29/2022	Worked on my version of scramble word game	11:00	1:00	2
1/29/2022	Worked on oscramble word game	3:00	4:00	1
			Total:	10.5
Week 4				
2/1/2022	Met with instructer and collaborated with team	5:00	6:30	1.5
2/2/2022	Worked on a version of the scramble game	6:30	7:30	1
2/3/2022	Collaborated with team	5:00	6:00	1
2/3/2022	Worked on prototype for scramble game	6:00	6:30	0.5
2/3/2022	Worked on prototype for scramble game	7:30	8:00	0.5
2/4/2022	Worked on prototype and design	12:00	2:00	2
2/4/2022	Tested Prototypes	1:00	3:00	2
2/5/2022	Tested Prototypes	3:00	4:30	1.5
			Total:	10

Week 5				
2/8/2022	Collaborated with team	5:00	6:00	1
2/9/2022	Worked on new prototytpes	3:00	4:00	1
2/10/2022	Worked on new prototypes	2:00	3:30	1.5
2/12/2022	Worked on prototypes and signage	11:00	3:00	4
2/12/2022	Worked on prototypes and signage	5:00	7:30	2.5
			Total:	10
Week 6				
2/13/2022	Worked on prototype and signage	12:00	4:00	4
2/14/2022	Worked on prototype and signage	4:00	8:00	4
			Total:	8
Week 7				
2/24/2022	Worked on signage and word scramble	12:00	4:30	4.5
2/24/2022	Collaborated with team	5:00	6:00	1
2/25/2022	Worked on Signage	11:00	1:00	2
2/26/2022	Worked on Signage and word scramble prototype	6:00	9:00	2.5
			Total:	10
Week 8				
3/1/2022	Collaborated with Hatch, Emily, and team	5:00	7:00	2
3/2/2022	Worked on signage	12:00	2:00	2
3/3/2022	Collaborated with team	4:30	5:30	1
3/3/2022	Worked on Native American Experience	5:30	6:30	1
3/4/2022	Worked on Native American Experience	12:00	2:00	2
3/5/2022	Worked on Native American Experience	3:00	5:00	2
			Total:	10
Week 9				
3/15/2022	Collaborated with team	5:15	6:15	1
3/15/2022	Worked on Native American Experience	6:15	7:15	1
3/17/2022	Worked on Native American Experience	1:30	3:30	2
3/17/2022	Collaborated with team	5:00	6:00	1
3/17/2022	Worked on Native American Experience	6:00	7:30	1.5
3/18/2022	Worked on Native American Experience	12:00	3:00	3
			Total:	9.5

Appendices

Time Logs - Jake C.

Week 10				
3/22/2022	Worked on Native American Experience	4:00	5:00	1
3/22/2022	Collaborated with Hatch and team	5:00	6:00	1
3/22/2022	Worked on experiences	6:00	7:00	1
3/24/2022	Refined Native American Experience	3:30	5:00	1.5
3/24/2022	Collaborated with team	5:00	6:00	1
3/25/2022	Worked Native American Experience	12:00	1:00	1
			Total:	6.5
Week 11				
3/29/2022	Collaborated with team and worked on signage	5:00	7:30	2.5
3/31/2022	Collaborated with team	5:00	6:00	1
4/1/2022	Worked on Signage and Native Experience	6:00	8:00	2
4/2/2022	Worked on Native American Experience	10:00	11:30	1.5
			Total:	7
Week 12				
4/5/2022	Worked on Signage and Native American Experience	3:00	5:00	2
4/5/2022	Collaborated with Hatch and team and worked on Experiences	5:00	7:00	2
4/6/2022	Worked on Signage	12:00	2:00	2
4/7/2022	Worked on prototype and testing for Native Experience	5:00	6:30	1.5
4/9/2022	Prototyped, Tested Native American Experience, worked on other designs	3:00	5:00	2
			Total:	9.5
Week 13				
4/12/2022	Collaborated with team and	5:00	7:00	2
4/14/2022	Collaborated with team	5:00	6:00	1
4/15/2022	Worked on new Native Experience	6:00	9:00	3
4/16/2022	Testing Experiences	11:00	2:00	3
			Total:	9
Week 14				
4/18/2019	Hiked time cave trail finding spots for signs	8:00	11:00	3
4/19/2022	Collaborated with Hatch and team and worked on new temporary Native Experience	5:00	6:00	1
4/20/2022	Worked on testing	2:00	4:00	2
4/21/2022	Worked on alternate Native American experienc	4:00	6:00	2
4/23/2022	Testing	8:00	10:00	2
			Total:	10

Jake Carter

Date	Description	Time In	Time Out	Hours
Week 1				
5/19/2022	Worked on "Home" page for all experiences	11:00	1:00	2
5/24/2022	Hiked Timp trail and took down old signs	5:30	7:30	2
5/25/2022	Brainstormed and tried new ideas for Native Experience	1:00	2:30	1.5
5/25/2022	Meeting with Harper and Team	4:00	4:30	1
			Total:	**6.5**
Week 2				
5/27/2022	Worked on Home page (Found image, worked on layout, and text)	12:00	2:00	2
5/28/2022	Worked on home page and researched and brainstormed alternative native experience	11:00	3:00	4
5/29/2022	Tested our current experiences	1:00	2:00	1
5/29/2022	Worked on Alternative Native Experience	5:00	7:00	2
			Total:	**9**
Week 3				
5/30/2022	Worked on Indigenous experience	11:00	2:00	3
6/1/2022	Worked on Home page and met with team	4:00	5:00	1
6/2/2022	Looked over and tested current experience	8:00	9:30	1.5
6/3/2022	Worked on Home Page	9:00	10:00	1
			Total:	**6.5**
Week 4				
6/12/2022	Worked on Native Experience	10:00	11:00	1
6/12/2022	Worked on Home page	10:00	12:00	2
6/12/2022	Worked on Native Experience	7:30	10:30	3
			Total:	6
Week 5				
6/16/2022	Worked on Native Experience	12:00	1:00	1.5
6/19/2022	Worked on Native Experience	8:00	9:30	1.5
			Total:	3

Appendices

Time Logs - Jake C.

Week 6				
6/20/2022	Added QR codes to signage and edited signage designs and indigenous experience	7:00	8:30	1.5
6/21/2022	Meeting and Timp Cave trail with Emily and Cami	8:00	10:30	2.5
6/22/2022	Worked on bird experience and met with Harper and team and started making feedback survey	2:00	5:00	3
			Total:	7

Week 7				
6/29/2022	Worked on sign for the first experience beta testing and survey	10:00	11:00	1
6/29/2022	Met with team and printed signs for testing on trail	4:00	5:00	1
6/30/2022	Set up solar paners and servers on the trail	8:00	5:00	9
			Total:	11

Week 8				
7/8/2022	Worked on bird experience	5:00	6:00	1
7/10/2022	Worked on Bird experience and looked over file sent by Dan	7:30	8:30	1
			Total:	2

Week 9				
7/10/2022	Looked over bird experience from preivous group and worked on current bird experience	11:00	12:00	1
7/11/2022	Worked on Bird Experience	9:30	11:00	1.5
7/13/2022	Met with team finished up design for the bird experience	4:00	4:30	0.5
			Total:	3

Week 10				
7/18/2022	Hiked Timp trail to test wifi connection	8:00	12:00	4
7/19/2022	Worked signage	9:00	9:30	0.5
7/20/2022	Meeting with group and Harper	4:00	5:00	1
7/22/2022	Printed signs Staples	3:00	4:00	1
			Total:	6.5

Week 11				
7/27/2022	Printed signs at UVU campus	10:00	11:30	1.5
7/27/2022	Meeting with team	4:00	4:15	0.25
7/28/2022	Delivered signs to Cami and the Timp Cave trail	10:15	11:45	1.75
7/30/2022	Hiked Timp Cave trail and made sure signs worked	8:00	12:30	4.5
			Total:	8

Week 12				
8/3/2022	Met with Harper and team	4:00	5:00	1
8/4/2022	Looked over designs and made sure there were not errors	11:00	11:30	0.5
			Total:	1.5

Week 13				
8/9/2022	Worked on design document	7:00	9:30	2.5
8/10/2022	Installed signage on the Timp Cave trail	8:00	1:00	5
8/11/2022	Worked on design document	11:00	2:25	3.25
8/12/2022	Worked on design document	12:00	1:30	1.5
			Total:	11.75